

```
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Loading LTX-Video model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils.http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
model_index.json 100% 412/412 [00:00<00:00, 50.8kB/s]
Download complete: 100% 28.4G/28.4G [01:17<00:00, 715MB/s]
```

Fetching 16 files: 100%  16/16 [01:17:00.00, 4.41s/s]
Loading pipeline components...: 100%  5/5 [00:46:00.00, 7.05s/s]
Loading checkpoint shards: 100%  2/2 [00:07:00.00, 3.59s/s]
Loading weights: 100%  219/219 [00:32:00.00, 3.32s/s]
LTX-Video model loaded successfully on L4.
GPU memory used: 14.25 GB
GPU memory available: 23.66 GB

```
#Generating first video on L4
import torch
from diffusers.utils import export_to_video
from datetime import datetime

prompt = "A woman with long brown hair smiling gently at the camera in a sunlit park, natural lighting, soft breeze moving her hair, cinematic quality"
negative_prompt = "blurry, low quality, distorted, unnatural"

generator = torch.Generator(device="cuda").manual_seed(42)

print(f"Prompts: {prompt}\n")
print("Generating video on L4 (should take 3-7 minutes)...")

start_time = datetime.now()

video_frames = pipe(
    prompt=prompt,
    negative_prompt=negative_prompt,
    width=704,
    height=480,
    num_frames=73,
    num_inference_steps=30,
    guidance_scale=3.0,
    generator=generator
).frames[0]

end_time = datetime.now()
duration = (end_time - start_time).total_seconds()

print(f"\nGeneration complete in (duration:.1f) seconds ((duration/60:.1f) minutes)")
print(f"Number of frames: {len(video_frames)}")
```

Prompt: A woman with long brown hair smiling gently at the camera in a sunlit park, natural lighting, soft breeze moving her hair, cinematic quality

Generating video on L4 (should take 3-7 minutes)...

100%  30/30 [00:24:00.00, 1.19H/s]

Generation complete in 27.7 seconds (0.5 minutes)
Number of frames: 73

```
#Saving video and metadata to Drive
from datetime import datetime
import json

timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
video_id = f"ltx_001_{timestamp}"

video_path = f'{DRIVE_PROJECT}/videos/generated/ltx/{video_id}.mp4'
metadata_path = f'{DRIVE_PROJECT}/metadata/{video_id}.json'

# Export the video
export_to_video(video_frames, video_path, fps=24)

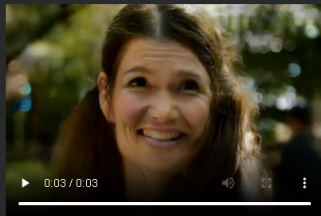
# Save metadata
metadata = {
    "video_id": video_id,
    "generator": "LTX-Video",
    "generator_version": "Lightricks/LTX-Video",
    "prompt": prompt,
    "negative_prompt": negative_prompt,
    "seed": 42,
    "width": 512,
    "height": 320,
    "num_frames": 49,
    "fps": 24,
    "num_inference_steps": 30,
    "guidance_scale": 3.0,
    "generation_time_seconds": duration,
    "generation_datetime": timestamp,
    "gpu_used": "T4",
    "optimizations": ["cpu_offload", "vae_slicing", "vae_tiling"]
}

with open(metadata_path, 'w') as f:
    json.dump(metadata, f, indent=2)

print(f"Video saved: {video_path}")
print(f"Metadata saved: {metadata_path}")
```

Video saved: /content/drive/MyDrive/msc_deepfake_project/videos/generated/ltx/ltx_001_20260716_133303.mp4
Metadata saved: /content/drive/MyDrive/msc_deepfake_project/metadata/ltx_001_20260716_133303.json

```
#View the generated video inline
from IPython.display import Video
Video(video_path, embed=True, width=400)
```



```
#Downloading the video to my local machine
from google.colab import files
files.download(video_path)
```

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 Variables  Terminal



✓ 2:34 PM